

The Implemented Algorithm Used to Find the Spikes in Data

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Introduction

For my web application, there is an algorithm made to track the spikes of the given data. This document explains this built algorithm and how it reliably tracks such spikes. Note that 'peaks' is used to refer to both valleys and peaks throughout this script, where lower peaks are valleys.

The Algorithm

The algorithm is built by hysteresis filtering the given data to remove all fluctuations in the data that are less than a given threshold. Thereafter, the peaks in the hysteresised data are found. The peaks sequentially are then spikes in the data that make clear the changes that are larger than the threshold.

Hysteresis Filtering

Hysteresis filtering is when small fluctuations in the data are removed [1]. Figure 1 presents an example of such.

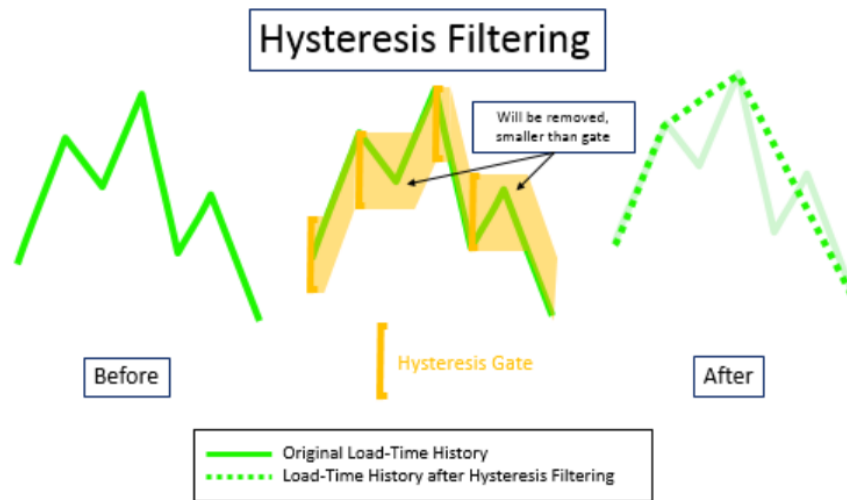


Figure 1: Hysteresis filtering, from [1]

As Figure 1 and Siemens' article explains, the filtering is done by defining a hysteresis gate. Any fluctuations in the data that are less than the hysteresis

gate are then removed from the data, as the figure's before and after load-time history demonstrates [1]. The value of the hysteresis gate is what was previously referred to as the threshold.

Peaks in Data

To find the peaks in the data, both for hysteresis filtering and for finding the peaks of the hysteresised data, SciPy's method for finding peaks was utilized [2]. Because SciPy's method for finding peaks does not consider edge data, the algorithm manually considers each edge also as peaks. Figure 2 illustrates where peaks are on a plot of data.

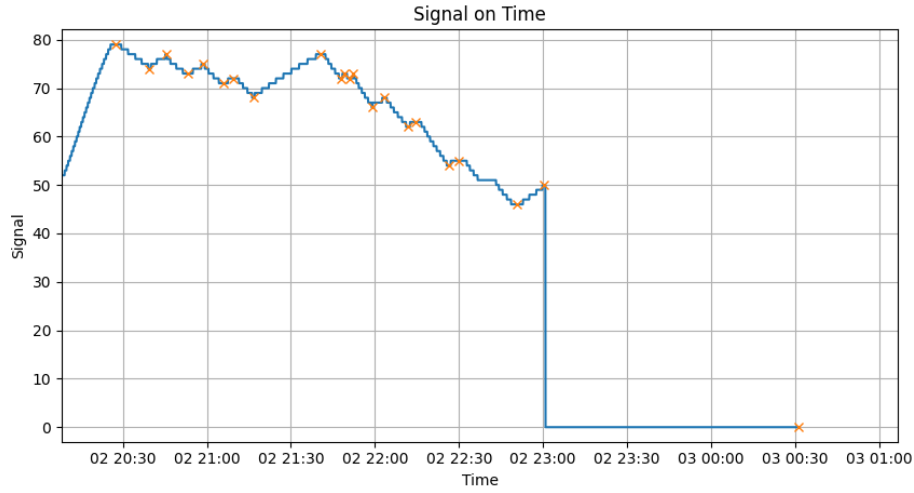


Figure 2: Peaks in data

The blue graph is the data itself, where the orange x-es represent the peaks. The plot shown is not all of the data and peaks, only the last portion. That is why the end of the graph is considered an edge (and therefore a peak), and not what optically seems like the beginning of the graph.

Visualization of the Algorithm

After finding the peaks of the hysteresised data and sequentially plotting them along with the data itself, one should get a plot that is along the lines of what Figure 3 shows. The chosen threshold was 5.0.

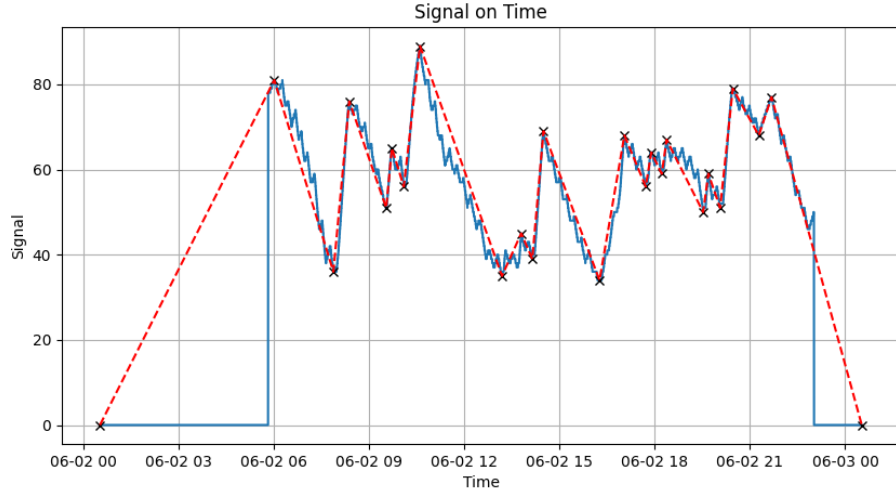


Figure 3: A plot of the data and the hysteresised data's peaks

The blue graph is the data itself, the black x-es the peaks of the hysteresised data, and the red dashed line a plot of the peaks, sequentially.

As the hysteresised data filtered out all fluctuations which are less than the threshold, the peaks which are found to said data now sequentially must have a distance larger than threshold. Therefore, plotting the peaks leads to a clear illustration of all changes in the data that are larger than threshold. To find out how large the spikes in the data are, the distance between the upper and lower peaks in the spike can simply be measured.

Conclusion

The algorithm made to track the spikes of the data functions by using Siemens' method of hysteresis filtering, to remove all fluctuations that are less than a chosen threshold [1]. Then, the peaks of the hysteresised data are found using SciPy's method to do such [2]. The peaks are then successively the changes in the data which are larger than the threshold.

References

- [1] Siemens, *Rainflow Counting*, 1st edition, 2019. Website [here](#).
- [2] SciPy, *scipy.signal.find_peaks*. Website [here](#)